LOW-RESOURCE HIGH-THROUGHPUT ENGINEERING

Boardgame 18M+ Rating Data Relational Inner Join Pipeline

Zero-Cloud OpEx relational data ingestion, cleaning, and persistence built under severe local hardware constraints in R.

DATASET VOLUME

18,000,000+ Records

PIPELINE ENGINE

R Tidyverse & readr

HARDWARE CONSTRAINTS

Target Environment & Machine Specs

Core i3 1.2GHz

Dual-Core Host CPU

4.0 GB RAM

Strict Physical RAM Ceiling

~2.0 GB OS Overhead

Windows 10 Base Footprint

No Cloud Support

Air-Gapped Local Execution

Available Process Budget

With Windows 10 overhead taking ~2.0 GB, **only ~2.0 GB RAM remains** to buffer, clean, join, and export 18M+ rows.

MEMORY GUARD ENABLED

SYSTEM ARCHITECTURE

Data Pipeline Stage Flowchart

1. Ingestion

Stream Raw CSVs via **readr C++ fast parser** into memory buffer.

2. Cleaning

Deduplicate & Cast data types with **non-zero ID filters**.

3. Compression

Vectorized Join via **BGGId primary key matching**.

4. Persistence

Disk Export of **pruned relational dataset** to CSV format.

PIPELINE STAGE 1

R Data Ingestion & Schema Inspection

Ingestion_Script.R — readr v2.1

```r
library(tidyverse)
library(readr)

# Ingest 18M+ Ratings raw CSV
data_rating ← read_csv(file.choose())
glimpse(data_rating)

# Ingest Boardgame Subcategory relational mapping
data_subcategory ← read_csv(file.choose())
glimpse(data_subcategory)
```

Fast C++ Parser

readr::read_csv utilizes multi-threaded C++ backends for instant disk-to-memory streaming.

Zero-Copy Glimpse

glimpse() inspects column types without triggering memory reallocation or buffer copies.

PIPELINE STAGE 2

Data Cleaning & Sanitization Logic

Cleaning_Logic.R — dplyr Pipe Engine

```r
# Sanitize Ratings: distinct, type coercion, non-zero filters
cleaned_data_rating ← data_rating %>%
  distinct() %>%
  mutate(
    BGGId = as.integer(BGGId),
    Rating = as.numeric(Rating)
  ) %>%
  filter(BGGId > 0 & Rating > 0)

# Sanitize Subcategory lookup table
cleaned_data_subcategory ← data_subcategory %>%
  distinct() %>%
  mutate(BGGId = as.integer(BGGId)) %>%
  filter(BGGId > 0)
```

De-duplication

distinct() strips duplicate submission rows to enforce absolute primary key integrity.

Boundary Safeguards

filter(BGGId > 0) purges corrupted IDs and invalid zero ratings before relational operations.

PIPELINE STAGE 3 & 4

Relational Inner Join & Persistence

Relational_Join.R — Inner Join Engine

```r
# Execute memory-optimized relational Inner Join
join_report ← cleaned_data_rating %>%
  inner_join(cleaned_data_subcategory, by = 'BGGId')

# Select target subcategories & export dataset
final_report ← join_report %>%
  select(BGGId, Rating, Username, Puzzle, Exploration, Electronic

write.csv(final_report, 'Boardgame_18M_rating_join_data.csv',
          row.names = FALSE)
```

Relational Inner Join

inner_join() merges 18M+ ratings with subcategories indexed strictly by BGGId key.

Selective Feature Export

select() projects essential metadata before disk persistence to maximize storage economy.

OPEN SOURCE REPOSITORY & PROFILE

Connect & Explore The Code

Access the complete **R pipeline codebase**, documentation, and operational benchmarks on GitHub or reach out on LinkedIn.

GitHub Repository www.github.com/talkmridul-bit

LinkedIn Profile www.linkedin.com/in/niaz-shoyab-495061b8